Supplementary material: trustworthiness is not one property in single-cell cervical cytology classification

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This supplement holds tables and figures referenced in the main manuscript as Tables S1–S14 and Figures S1–S7. Section numbers refer to the main manuscript.

Table S1: In-distribution benchmark on SIPaKMeD, all 12 models, sorted by balanced accuracy (seed 13).

Model	Bal. acc.	Macro F1	AUROC	Params
EfficientViT-M0	**0.9654**	**0.9656**	0.9985	**2,157,401**
EfficientViT-B0	0.9623	0.9624	0.9983	2,135,413
DeiT-Tiny	0.9606	0.9607	0.9974	5,525,381
Swin-Tiny	0.9605	0.9606	0.9965	27,523,199
TinyViT-5M	0.9587	0.9589	0.9979	5,073,369
MobileViT-S	0.9570	0.9575	0.9979	4,940,837
ResNet-50	0.9542	0.9540	0.9966	23,518,277
LeViT-128	0.9458	0.9456	0.9946	8,447,786
MobileViTv2	0.9387	0.9401	0.9966	4,391,406
EfficientNet-Lite0	0.9375	0.9376	0.9944	3,377,413
ViT-Small	0.9374	0.9380	0.9960	21,667,589
MobileNetV2	0.9357	0.9365	0.9952	2,230,277

Table S2: Balanced accuracy of all 12 models, seed 13, sorted by Mendeley LBC score. The SIPaKMeD column is five-class in-distribution balanced accuracy from the same run (chance 0.20); the Herlev and Mendeley LBC columns are zero-shot binary normal/abnormal balanced accuracy (chance 0.50). The scales are not comparable, so only ranks are compared across columns.

Model (family, params)	SIPaKMeD (5-class)	Herlev	Mendeley LBC
EfficientNet-Lite0 (CNN, 3.38M)	0.9455	0.5970	**0.6674**
ResNet-50 (CNN, 23.5M)	0.9556	0.6838	0.6082
MobileViTv2 (eff. ViT, 4.39M)	0.9638	**0.7501**	0.5418
MobileNetV2 (CNN, 2.23M)	0.9505	0.6919	0.5391
DeiT-Tiny (eff. ViT, 5.5M)	0.9504	0.7151	0.5230
EfficientViT-B0 (eff. ViT, 2.14M)	0.9738	0.6818	0.5217
Swin-Tiny (full ViT, 27.5M)	0.9590	0.7282	0.5192
ViT-Small (full ViT, 21.7M)	0.9573	0.7049	0.5084
LeViT-128 (eff. ViT, 8.4M)	0.9526	0.5777	0.4764
EfficientViT-M0 (eff. ViT, 2.16M)	0.9737	0.6716	0.4731
TinyViT-5M (eff. ViT, 5.1M)	0.9685	0.7327	0.4543
MobileViT-S (eff. ViT, 4.94M)	0.9520	0.7104	**0.2927** (AUROC 0.215)

Table S3: Zero-shot binary evaluation of all 12 models on APCEM, sorted by AUROC. This table comes from a separate training run, so the five-class in-distribution balanced accuracy in the SIPaKMeD column differs from Table S2.

Model (family, params)	SIPaKMeD (5-class)	APCEM bal. acc.	APCEM AUROC
MobileNetV2 (CNN, 2.23M)	0.9623	**0.6158**	**0.6528**
EfficientNet-Lite0 (CNN, 3.38M)	0.9309	0.5393	0.5701
LeViT-128 (eff. ViT, 8.4M)	0.9001	0.5263	0.5376
Swin-Tiny (full ViT, 27.5M)	0.9330	0.4724	0.3082
EfficientViT-B0 (eff. ViT, 2.14M)	0.9638	0.4481	0.3206
DeiT-Tiny (eff. ViT, 5.5M)	0.9589	0.4469	0.3799
MobileViTv2 (eff. ViT, 4.39M)	0.9570	0.4290	0.4120
TinyViT-5M (eff. ViT, 5.1M)	0.9670	0.4100	0.3681
EfficientViT-M0 (eff. ViT, 2.16M)	0.9621	0.3730	0.3407
MobileViT-S (eff. ViT, 4.94M)	0.9687	0.3621	0.3161
ViT-Small (full ViT, 21.7M)	0.9538	0.3444	0.2555
ResNet-50 (CNN, 23.5M)	0.9639	**0.3339**	0.2929

Table S4: Mean ± standard deviation of balanced accuracy over three training seeds (13, 17, 29), in the row order of Table S2. The SIPaKMeD column is five-class; the external columns are binary. Seed 13 values are those of Tables S2 and S3, which come from separate runs.

Model	SIPaKMeD (5-class)	Herlev	Mendeley LBC	APCEM
EfficientNet-Lite0	0.951 ± 0.006	0.630 ± 0.030	0.630 ± 0.106	0.544 ± 0.027
ResNet-50	0.957 ± 0.002	0.689 ± 0.005	0.537 ± 0.062	0.390 ± 0.072
MobileViTv2	0.964 ± 0.004	0.721 ± 0.025	0.516 ± 0.064	0.466 ± 0.033
MobileNetV2	0.946 ± 0.004	0.659 ± 0.030	0.563 ± 0.083	0.586 ± 0.026
DeiT-Tiny	0.955 ± 0.004	0.700 ± 0.015	0.484 ± 0.036	0.385 ± 0.059
EfficientViT-B0	0.966 ± 0.008	0.667 ± 0.017	0.575 ± 0.047	0.442 ± 0.035
Swin-Tiny	0.961 ± 0.005	0.720 ± 0.009	0.481 ± 0.034	0.419 ± 0.047
ViT-Small	0.957 ± 0.008	0.705 ± 0.048	0.477 ± 0.055	0.392 ± 0.045
LeViT-128	0.900 ± 0.065	0.538 ± 0.034	0.507 ± 0.031	0.518 ± 0.016
EfficientViT-M0	0.966 ± 0.009	0.681 ± 0.015	0.502 ± 0.038	0.409 ± 0.043
TinyViT-5M	0.968 ± 0.001	0.706 ± 0.025	0.453 ± 0.010	0.422 ± 0.018
MobileViT-S	0.958 ± 0.005	0.689 ± 0.029	0.387 ± 0.084	0.372 ± 0.024

Table S5: Parameters, FLOPs (multiply-add counted as 2), and median single-image CPU latency for all 12 models (untrained weights, Kaggle GPU T4 x2 notebook CPU, 4 vCPUs).

Model	Params	FLOPs (G)	CPU latency (ms)
EfficientViT-M0	2,157,401	0.158	36.2
EfficientViT-B0	2,135,413	0.198	41.6
MobileNetV2	2,230,277	0.599	33.5
EfficientNet-Lite0	3,377,413	0.768	**137.7**
LeViT-128	8,447,786	0.809	17.5
DeiT-Tiny	5,525,381	2.149	**15.4**
TinyViT-5M	5,073,369	2.510	44.9
MobileViT-S	4,940,837	3.645	66.3
MobileViTv2	4,391,406	3.623	65.2
ResNet-50	23,518,277	8.174	32.0
ViT-Small	21,667,589	8.482	26.8
Swin-Tiny	27,523,199	8.980	54.0

Table S6: Explainability scores on all 596 cluster-grouped test cells, mean [95% cluster-bootstrap interval]. Lower is better for deletion AUC, and higher is better for insertion AUC and nucleus concentration (1.0 is chance).

Backbone	Method	Deletion AUC	Insertion AUC	Nucleus concentration
DeiT-Tiny	Relevancy propagation	0.727 [0.678, 0.777]	0.966 [0.957, 0.973]	3.79 [2.80, 4.81]
DeiT-Tiny	Attention rollout	0.807 [0.768, 0.845]	0.941 [0.925, 0.955]	1.35 [1.15, 1.59]
DeiT-Tiny	Grad-CAM	0.816 [0.779, 0.852]	0.904 [0.876, 0.931]	1.19 [1.01, 1.37]
ViT-Small	Relevancy propagation	0.721 [0.653, 0.789]	0.961 [0.952, 0.970]	4.79 [3.49, 6.17]
ViT-Small	Attention rollout	0.766 [0.699, 0.831]	0.947 [0.932, 0.959]	1.95 [1.69, 2.22]
ViT-Small	Grad-CAM	0.910 [0.886, 0.931]	0.780 [0.714, 0.843]	1.17 [0.91, 1.43]

Table S7: Explainability method comparison on DeiT-Tiny, 50 SIPaKMeD test images (10 per class), all matched to a real nucleus polygon.

Method	Deletion AUC↓	Insertion AUC↑	Nucleus conc.↑
Relevancy propagation	**0.734**	**0.970**	**2.714**
Grad-CAM (ViT-adapted)	0.818	0.877	1.472
Attention rollout	0.825	0.936	1.177

Table S8: Explainability ranking on a second backbone (ViT-Small) versus the original (DeiT-Tiny).

	DeiT-Tiny (original)			ViT-Small (2nd backbone)		
Method	Del.↓	Ins.↑	Conc.↑	Del.↓	Ins.↑	Conc.↑
Relevancy prop.	**0.734**	**0.970**	**2.71**	**0.816**	**0.965**	**4.20**
Grad-CAM	0.818	0.877	1.47	0.888	0.895	1.35
Attn. rollout	0.825	0.936	1.18	0.888	0.927	0.97

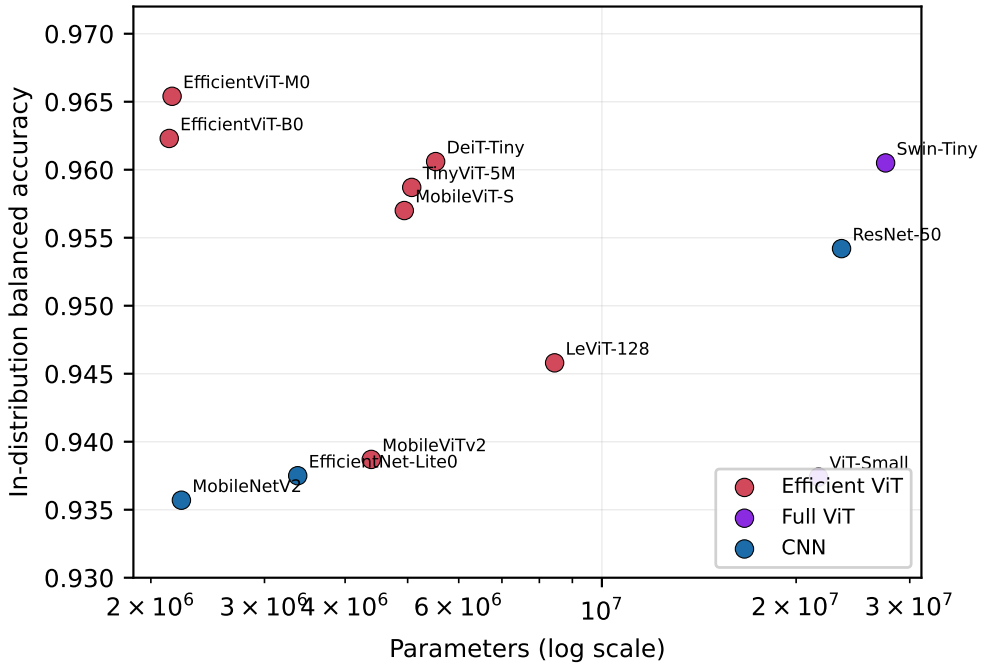


Figure S1: Original cell-level split, seed 13. Parameters (log scale) versus in-distribution balanced accuracy, colored by architecture family. The two efficient-ViT models with the fewest parameters (EfficientViT-M0 and -B0) are also the two most accurate in this run.

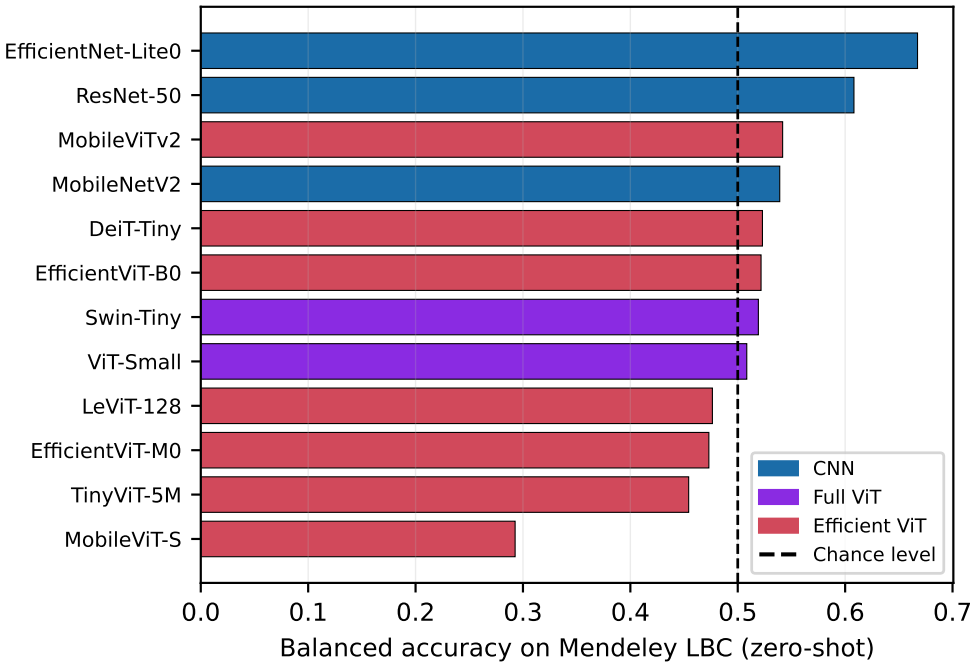


Figure S2: Original cell-level split, seed 13. Balanced accuracy on Mendeley LBC (zero-shot), sorted and colored by architecture family. All three CNNs cluster above chance, and eight of nine transformer-family models cluster at or below chance, with MobileViT-S the sole catastrophic outlier.

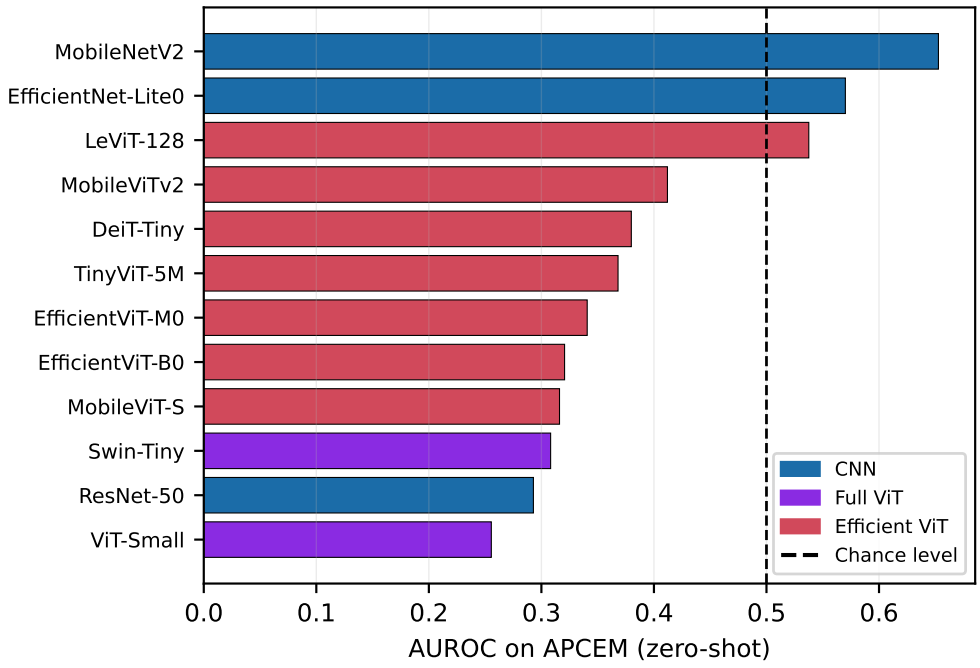


Figure S3: Original cell-level split, seed 13. AUROC on APCEM (zero-shot), sorted and colored by architecture family. Unlike Mendeley LBC, the CNN family spans both the best (MobileNetV2) and one of the worst (ResNet-50) models; no family-level pattern survives this third dataset.

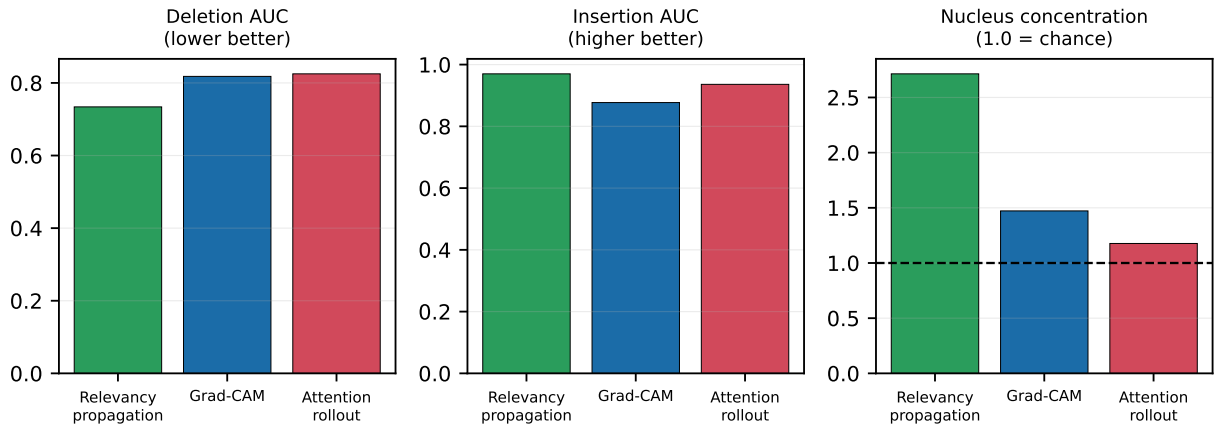


Figure S4: Deletion AUC, insertion AUC, and nucleus concentration ratio for the three explainability methods. Relevancy propagation wins on all three, with the widest margin on nucleus concentration.

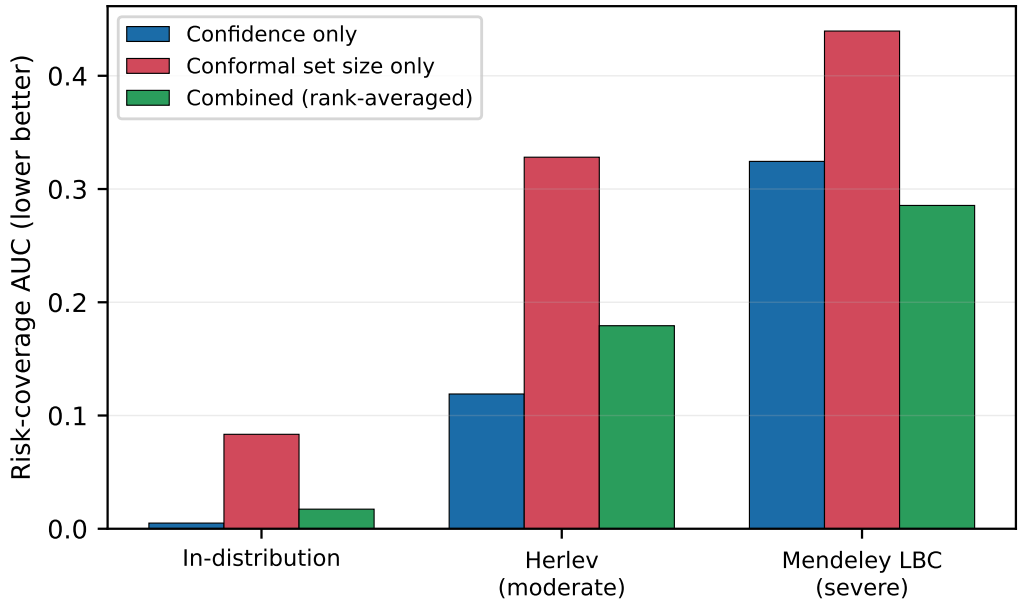


Figure S5: Risk-coverage AUC for confidence alone, conformal set size alone, and the rank-averaged combination at three conditions (original cell-level run). Combining hurts in distribution and under moderate shift but helps under severe shift.

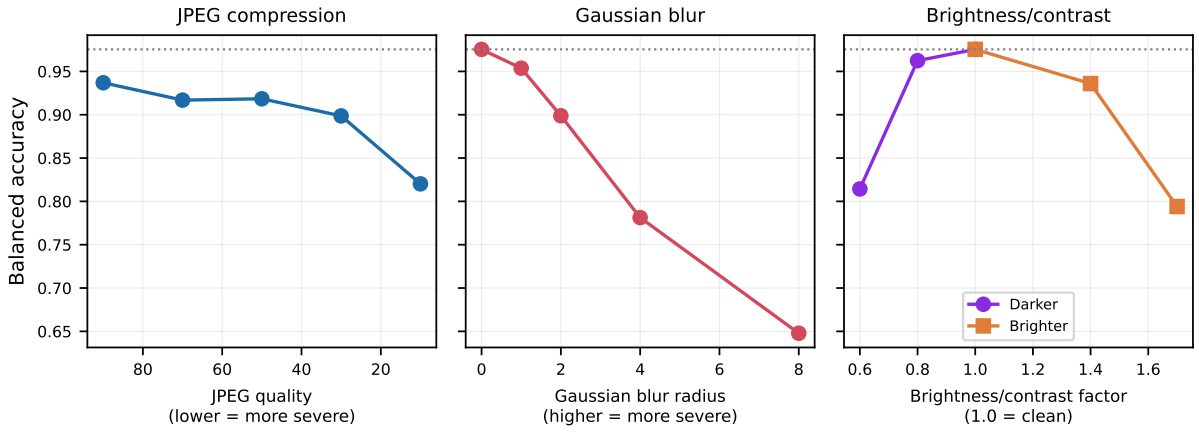


Figure S6: Original cell-level split. Balanced accuracy versus corruption severity for three synthetic degradations (EfficientViT-M0, SIPaKMeD test set). The dotted line marks the clean baseline (0.975). Gaussian blur causes the steepest decline.

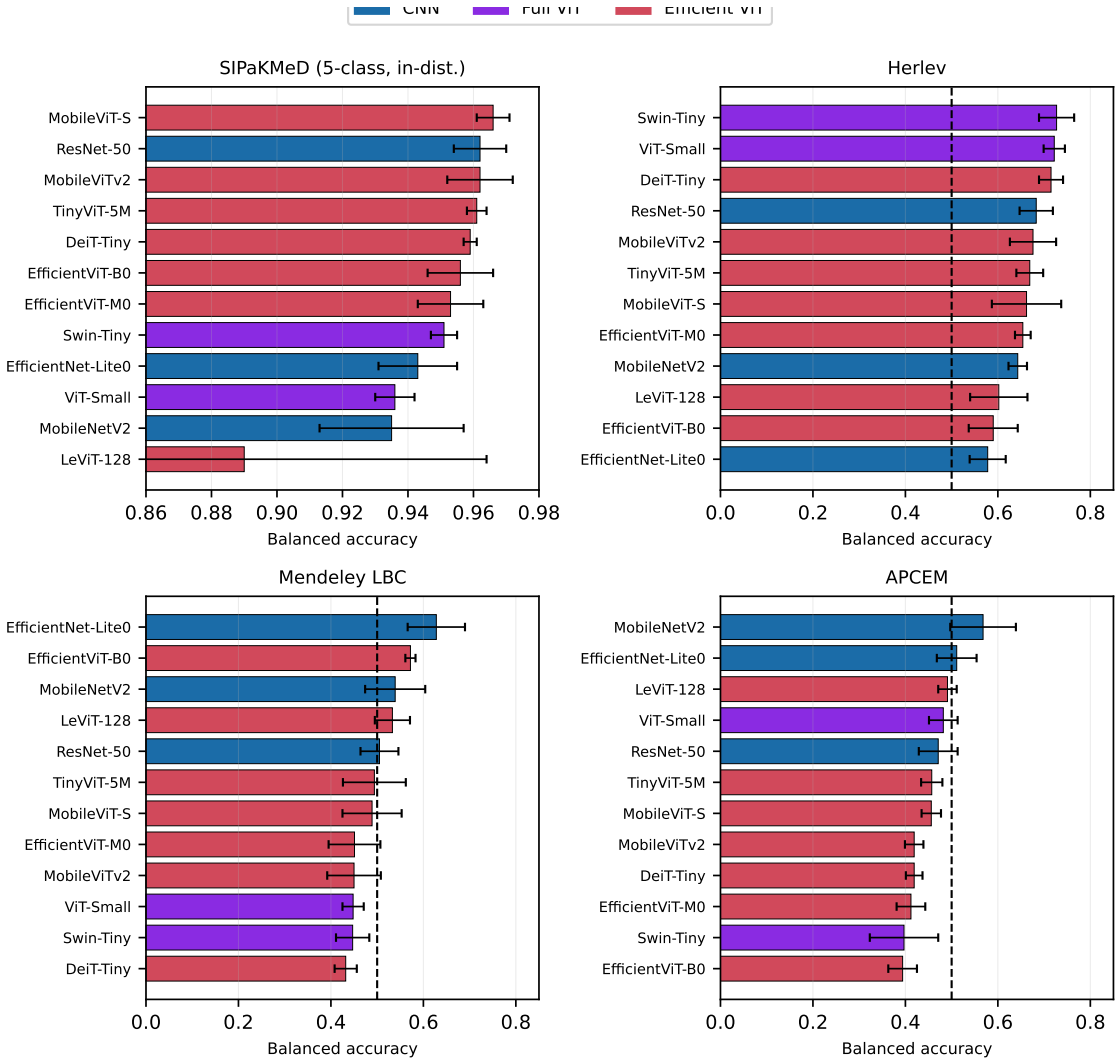


Figure S7: Cluster-grouped split, seeds 13/17/29. Balanced accuracy, mean ± standard deviation over three training seeds, same fixed cluster-grouped split at every seed; same data as Table S12, sorted by mean within each panel. The in-distribution ranking reorders across seeds more than the three external-dataset leaders do (Sec. 4.2 in the main text).

Table S9: Class-by-class binary label mapping and image counts. NILM: negative for intraepithelial lesion or malignancy; ASC-US: atypical squamous cells of undetermined significance; ASC-H: atypical squamous cells, cannot exclude a high-grade lesion; LSIL and HSIL: low-grade and high-grade squamous intraepithelial lesion; SCC: squamous cell carcinoma. Metaplastic cells in SIPaKMeD are benign and mapped to normal; the equivocal ASC-US and ASC-H categories are grouped with abnormal, a convention that is itself debatable.

Dataset	Binary label	Original classes (images)
SIPaKMeD (4,049)	Normal (2,411)	Superficial-intermediate (831), parabasal (787), metaplastic (793)
	Abnormal (1,638)	Koilocytotic (825), dyskeratotic (813)
Herlev (917)	Normal (242)	Normal columnar (98), normal intermediate (70), normal superficial (74)
	Abnormal (675)	Light dysplastic (182), moderate dysplastic (146), severe dysplastic (197), carcinoma in situ (150)
Mendeley LBC (962)	Normal (612)	NILM (612)
	Abnormal (350)	LSIL (113), HSIL (163), SCC (74)
APCEM (4,253)	Normal (779)	NILM (779)
	Abnormal (3,474)	ASC-US (340), ASC-H (119), LSIL (1,241), HSIL (1,300), SSC (474)

Table S10: Reproducibility information. Items marked "not recorded" could not be recovered from the repository.

Component	Information
Software versions	All cluster-grouped-split Kaggle sessions (benchmark re-run, seed check, uncertainty-quantification comparison, symmetric ensemble check): Python 3.13.15, torch 2.11.0+cu128, timm 1.0.29, numpy 2.1.3, CUDA 12.8, cuDNN 9.19, identical across sessions. Original cell-level runs: minimum versions only (torch $\geq$2.2, timm $\geq$1.0); exact versions not recorded
Training hardware	Google Colab T4 GPU (original cell-level benchmark; other cell-level stages not individually recorded); Kaggle GPU T4 x2 notebook for all cluster-grouped-split work (benchmark re-run: 12 models in 1.6 GPU hours; seed check, uncertainty-quantification comparison, and symmetric ensemble check: individual session durations not recorded)
Latency hardware	Kaggle GPU T4 x2 notebook CPU, 4 vCPUs (Kaggle documents the underlying Intel platform as variable across Skylake, Broadwell, and AMD rather than a single pinned model); explanation-cost pair measured on the same platform
Random seeds	Cell-level and cluster-grouped benchmarks: training seeds 13 (all stages), 17 and 29 (seed check). Uncertainty-quantification comparison (EfficientViT-M0, cluster-grouped split): baseline/MC Dropout/evidential trained at seed 13; MC Dropout additionally retrained at seeds 17 and 29 for the significance test (Table S13); deep ensemble built from three independently-seeded 5-member groups, 1–5, 6–10, and 11–15 (Table 8 in the main text). Split seed 13 plus class index for both split types.
Determinism	Only the PyTorch seed is set; deterministic cuDNN not enabled; NumPy, Python, and data-loader workers not separately seeded
Data split	Two split types, both stratified by class, both identical across training seeds within that type (main text, Sec. 3.3): the *cell-level* manifest (single cells shuffled, 87.8% test-train source-image overlap) and the *cluster-grouped* manifest (every source cluster image confined to one split, 2,439/630/384/596 train/val/calibration/test cells, 163 test clusters), the latter used as the primary benchmark and for the uncertainty-quantification comparison.
Checkpoint selection	Lowest validation loss, early stopping (patience 7, at most 30 epochs)

Table S11: Paired differences in balanced accuracy (percentage points, leader minus model) between the leader on each cluster-grouped dataset (named in the column header) and every other model, with 95% bootstrap intervals (2,000 resamples, matching the resampling count used elsewhere in this supplement; clusters resampled in distribution, images externally) and an asterisk when the interval excludes zero. Each model is one training run, so intervals capture test-sample uncertainty, not training variability.

Model	In-dist. (MobileViTv2)	Herlev (Swin-Tiny)	Mendeley LBC (EfficientNet-Lite0)	APCEM (MobileNetV2)
MobileViTv2	leader	+2.4 [+0.3, +4.6]*	+14.4 [+10.5, +18.4]*	+23.9 [+21.6, +26.3]*
MobileViT-S	+0.5 [-0.9, +1.8]	+8.8 [+6.2, +11.6]*	+21.6 [+17.4, +26.0]*	+17.2 [+15.2, +19.2]*
ResNet-50	+0.7 [-0.8, +2.1]	+2.5 [+0.2, +4.8]*	+9.1 [+5.8, +12.4]*	+13.3 [+10.7, +16.0]*
DeiT-Tiny	+1.0 [-0.5, +2.5]	+1.7 [-0.9, +4.5]	+17.2 [+14.1, +20.4]*	+21.4 [+18.8, +24.1]*
TinyViT-5M	+1.0 [-0.4, +2.4]	+4.4 [+2.0, +6.8]*	+21.4 [+17.6, +25.0]*	+21.4 [+19.4, +23.4]*
EfficientNet-Lite0	+1.5 [-0.5, +3.5]	+20.7 [+16.5, +24.7]*	leader	+12.8 [+10.7, +15.2]*
EfficientViT-B0	+1.5 [-0.3, +3.2]	+12.1 [+9.5, +14.8]*	+7.2 [+4.5, +10.0]*	+22.3 [+19.6, +25.0]*
Swin-Tiny	+1.7 [-0.2, +3.9]	leader	+19.6 [+16.1, +23.1]*	+31.6 [+29.3, +34.0]*
EfficientViT-M0	+2.2 [+0.3, +4.0]*	+9.1 [+6.4, +11.8]*	+18.4 [+15.0, +21.7]*	+20.7 [+18.2, +23.3]*
ViT-Small	+3.3 [+1.4, +5.5]*	+3.2 [+0.3, +6.0]*	+18.9 [+15.2, +22.4]*	+15.0 [+13.2, +16.9]*
MobileNetV2	+5.6 [+3.1, +8.2]*	+11.4 [+8.2, +14.6]*	+10.1 [+6.2, +14.0]*	leader
LeViT-128	+16.2 [+11.6, +21.6]*	+20.9 [+17.5, +24.3]*	+12.8 [+9.5, +15.9]*	+13.9 [+12.0, +15.7]*

Table S12: Cluster-grouped benchmark, mean $\pm$ standard deviation of balanced accuracy over three training seeds (13, 17, 29), identical cluster-grouped split held fixed across seeds. The SIPaKMeD column is five-class; the external columns are binary. Seed 13 values are those of the cluster-grouped benchmark table in the main text; rows are in the same order.

Model	SIPaKMeD (5-class)	Herlev	Mendeley LBC	APCEM
EfficientNet-Lite0	0.943 ± 0.012	0.578 ± 0.039	0.628 ± 0.062	0.511 ± 0.043
ResNet-50	0.962 ± 0.008	0.683 ± 0.036	0.505 ± 0.041	0.471 ± 0.042
MobileViTv2	0.962 ± 0.010	0.676 ± 0.050	0.450 ± 0.058	0.419 ± 0.020
MobileNetV2	0.935 ± 0.022	0.643 ± 0.020	0.539 ± 0.065	0.568 ± 0.071
DeiT-Tiny	0.959 ± 0.002	0.715 ± 0.026	0.432 ± 0.024	0.419 ± 0.018
EfficientViT-B0	0.956 ± 0.010	0.590 ± 0.053	0.572 ± 0.011	0.394 ± 0.031
Swin-Tiny	0.951 ± 0.004	0.727 ± 0.038	0.447 ± 0.036	0.397 ± 0.074
ViT-Small	0.936 ± 0.006	0.722 ± 0.023	0.448 ± 0.023	0.482 ± 0.031
LeViT-128	0.890 ± 0.074	0.602 ± 0.062	0.533 ± 0.038	0.491 ± 0.020
EfficientViT-M0	0.953 ± 0.010	0.654 ± 0.017	0.451 ± 0.056	0.412 ± 0.031
TinyViT-5M	0.961 ± 0.003	0.669 ± 0.029	0.494 ± 0.068	0.457 ± 0.023
MobileViT-S	0.966 ± 0.005	0.662 ± 0.075	0.489 ± 0.064	0.456 ± 0.021

Table S13: MC Dropout versus the deep ensemble, paired bootstrap test (2,000 resamples, matching the resampling count used elsewhere in this supplement) on risk-coverage AUC, EfficientViT-M0, cluster-grouped split. The ensemble is a single fixed set of 5 seed-13-trained members, reused at every row; MC Dropout is retrained fresh at each listed seed. Δ is ensemble RC-AUC minus MC Dropout RC-AUC (positive: MC Dropout better; negative: ensemble better). p is a two-sided bootstrap p-value computed directly from the resampled difference distribution (twice the smaller tail proportion, not inferred from the confidence-interval endpoints); p_{Holm} is the Holm step-down-adjusted p-value across all 9 tests in this table. Significant at $\alpha = 0.05$ after correction: marked with †.

Dataset	MC seed	MC RC-AUC	Ensemble RC-AUC	Δ	p	p_{Holm}	Sig. (Holm)
Herlev	13	0.2152	0.2234	+0.0083	0.549	0.549	
Herlev	17	0.2596	0.2234	−0.0362	0.0010	0.0060	†
Herlev	29	0.2406	0.2234	−0.0171	0.0780	0.1560	
Mendeley LBC	13	0.4594	0.3868	−0.0727	0.0010	0.0060	†
Mendeley LBC	17	0.3275	0.3868	+0.0593	0.0005	0.0045	†
Mendeley LBC	29	0.4304	0.3868	−0.0436	0.0080	0.0320	†
APCEM	13	0.7803	0.8206	+0.0403	0.0005	0.0045	†
APCEM	17	0.8320	0.8206	−0.0115	0.0350	0.1050	
APCEM	29	0.7736	0.8206	+0.0470	0.0005	0.0045	†

Table S14: Cluster-grouped benchmark, individual balanced accuracy at each training seed (13/17/29, slash-separated), underlying the means in Table S12. Same fixed cluster-grouped split at every seed; rows in the same order as Table S12.

Model	SIPaKMeD (5-class)	Herlev	Mendeley LBC	APCEM
MobileViTv2	0.968/0.967/0.950	0.722/0.623/0.683	0.487/0.383/0.480	0.410/0.441/0.405
MobileViT-S	0.963/0.972/0.962	0.659/0.589/0.739	0.416/0.513/0.537	0.476/0.434/0.459
ResNet-50	0.961/0.970/0.955	0.721/0.651/0.677	0.541/0.461/0.513	0.516/0.464/0.433
DeiT-Tiny	0.958/0.958/0.961	0.730/0.685/0.731	0.460/0.417/0.419	0.434/0.399/0.424
TinyViT-5M	0.958/0.963/0.960	0.703/0.656/0.649	0.418/0.512/0.551	0.434/0.459/0.479
EfficientNet-Lite0	0.953/0.930/0.945	0.540/0.618/0.575	0.632/0.688/0.563	0.520/0.548/0.464
EfficientViT-B0	0.953/0.948/0.968	0.625/0.528/0.616	0.559/0.581/0.576	0.426/0.392/0.364
Swin-Tiny	0.951/0.955/0.946	0.747/0.750/0.682	0.436/0.417/0.486	0.332/0.380/0.478
EfficientViT-M0	0.947/0.965/0.948	0.655/0.670/0.637	0.448/0.397/0.508	0.441/0.380/0.417
ViT-Small	0.935/0.931/0.943	0.715/0.748/0.703	0.443/0.427/0.473	0.498/0.501/0.446
MobileNetV2	0.912/0.935/0.957	0.633/0.666/0.630	0.531/0.479/0.608	0.648/0.541/0.514
LeViT-128	0.807/0.920/0.945	0.538/0.605/0.662	0.504/0.577/0.518	0.510/0.470/0.493